Question:
A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.
Answer Choices:
A. 10 % Zn
B. 22 % Zn
C. 20 % Zn
D. 19 % Zn
E. 15 % Zn
F. 18 % Zn
G. 16 % Zn
H. 14 % Zn
I. 12 % Zn
J. 25 % Zn

Reasoning + Answer:

<reasoning process here>
We are given:

- A 2 gram sample of rock yields 0.969 grams of ZnâĤĤFe(CN)âĤĨ.
- We are to find the percentage of Zn in the original rock sample.
- The reactions:
1. ZnS + 2HCl âĨĴ ZnClâĤĤ + HâĤĤS
2. 2ZnClâĤĤ + KâĤĦFe(CN)âĤĨ âĨĴ ZnâĤĤFe(CN)âĤĨ + 4KCl

We are told that the Zn in the rock is converted to ZnâĤĤFe(CN)âĤĨ via the above reactions.

We are given:
- Molecular weight of ZnâĤĤFe(CN)âĤĨ = 342.6 g/mol
- Atomic weight of Zn = 65.4 g/mol
- The precipitate is ZnâĤĤFe(CN)âĤĨ, which contains 2 Zn atoms.

So, in 1 mole of ZnâĤĤFe(CN)âĤĨ, there are 2 moles of Zn.

Step 1: Find the mass of Zn in 0.969 g of ZnâĤĤFe(CN)âĤĨ.

Mass of Zn in the precipitate = (2 ÃĹ atomic mass of Zn / molecular mass of ZnâĤĤFe(CN)âĤĨ) ÃĹ mass of precipitate

= (2 ÃĹ 65.4 / 342.6) ÃĹ 0.969

Calculate:

2 ÃĹ 65.4 = 130.8
130.8 / 342.6 âīĪ 0.3818

0.3818 ÃĹ 0.969 âīĪ ?

0.3818 ÃĹ 0.969 =
= 0.3818 ÃĹ (1 - 0.031) âīĪ 0.3818 - (0.3818 ÃĹ 0.031)
= 0.3818 - 0.011836 âīĪ 0.370

Better to compute directly:

0.3818 ÃĹ 0.969
= (0.38 ÃĹ 0.969) = 0.36822
(0.0018 ÃĹ 0.969) âīĪ 0.00174
Total âīĪ 0.36996 âīĪ 0.370 g

So, mass of Zn in the precipitate âīĪ 0.370 g

This Zn came from the original 2 g sample.

Therefore, % Zn in the rock = (mass of Zn / mass of sample) ÃĹ 100
= (0.370 / 2) ÃĹ 100 = 18.5%

Closest option to 18.5% is:

F. 18% Zn

But letâĢĻs compute more precisely.

Precise calculation:

(2 ÃĹ 65.4) = 130.8
130.8 / 342.6 = ?

130.8 Ã· 342.6 = ?